

Area of Rectangles

Time limit: 1.00 s

Memory limit: 512 MB

Given n rectangles, your task is to determine the total area of their union.

Input

The first line has an integer n : the number of rectangles.

After that, there are n lines describing the rectangles. Each line has four integers x_1, y_1, x_2 and y_2 : a rectangle begins at point (x_1, y_1) and ends at point (x_2, y_2) .

Output

Print the total area covered by the rectangles.

Constraints

- $1 \leq n \leq 10^5$
- $-10^6 \leq x_1 < x_2 \leq 10^6$
- $-10^6 \leq y_1 < y_2 \leq 10^6$

Example

Input	Output
3 1 3 4 5 3 1 7 4 5 3 8 6	24